

general feedback

- **don't show summaries, only show what I need to see – please only use *head***
 - or extract only the necessary bits: `summary(modelC1)$coefficients["attitudepol", "Std. Error"]`
- Remember to include the code that would give you your result
- when plotting, plot the model, not smoothed lines
- pdf's, not html's
- please don't answer in comments
- Remember to use the natural units, e.g. for freq it is Hz
- Plot meaningful x-axes and y-axes
- no interaction needed for no-pooling model, you shouldn't model slopes
- make sure you use (n-p) correction for variance – the *sigma* function will do this
 - and make sure to report residual variance and not sigma
- for pdf's make sure that you stay within margins
- don't remove the missing values – this is also information about the data

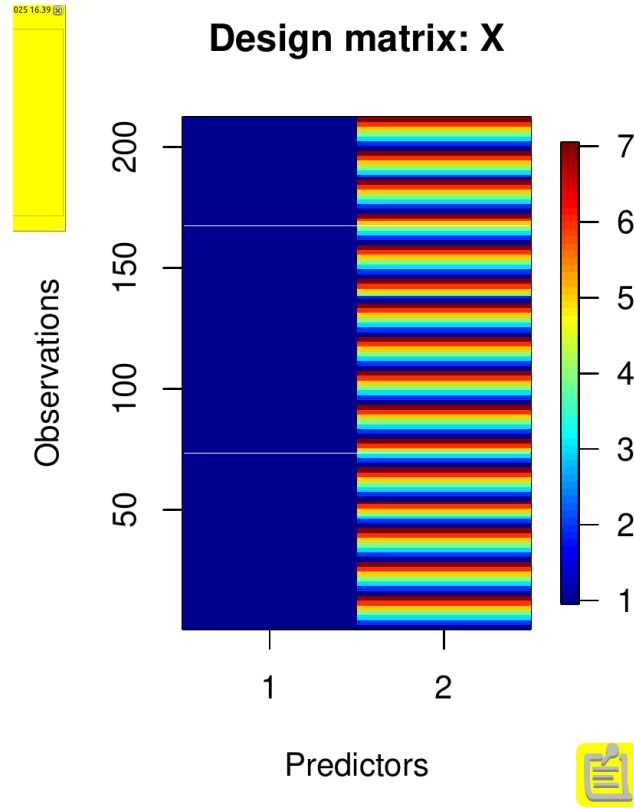
Remember to include the code that would give you your result

ii. include number of observations and missing observations

There are 224 observations in total - 16 different participants, 9 female, 7 male. There are observations of f0mn.



Meaningful axes



Don't flood the report

```
lapply(partial_pooling_comb_head, head, 4)
```

```
## $`subject:scenario`  
##      (Intercept)  
## F1:1  0.7169462  
## F1:2  0.1930955  
## F1:3  0.3967191  
## F1:4  2.2371162  
##  
## $subject  
##      (Intercept)  
## F1   -3.908901  
## F2   -5.391665  
## F3   -2.471632  
## F4    2.512939  
##  
## $scenario  
##      (Intercept)  
## 1    -5.25227  
## 2    1.000776
```



please don't answer in comments

```
## [1] 137.0633
```



```
# no_pooling and part_pooling are not at explaining the most variance.
```

```
# ii
```

margin problems (pdf)

```
#iv  
model_subject_scenario <- lmer(total_duration ~ f0mn + (1|subject) + (1|scenario) + (1|subject:scenario)
```

too much info

```
summary(no_pooling)
```

```
##
## Call:
## lm(formula = total_duration ~ f0mn + factor(scenario), data = politeness)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -26.041  -4.906  -2.007   2.397  52.992
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    28.02379     3.00851   9.315 < 2e-16 ***
## f0mn           -0.04476     0.01172  -3.820 0.000177 ***
## factor(scenario)2 -7.41634     3.01234  -2.462 0.014646 *
## factor(scenario)3 -8.22516     3.03978  -2.706 0.007390 **
## factor(scenario)4 -11.91661     3.04294  -3.916 0.000123 ***
## factor(scenario)5 -10.79057     2.97794  -3.624 0.000367 ***
## factor(scenario)6  31.06480     2.97389  10.446 < 2e-16 ***
## factor(scenario)7  44.11652     2.97450  14.832 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.71 on 204 degrees of freedom
## (12 observations deleted due to missingness)
## Multiple R-squared:  0.7812, Adjusted R-squared:  0.7737
## F-statistic: 104.1 on 7 and 204 DF, p-value: < 2.2e-16
```



I asked for
residual
variance
and I get
all this?

nice and concise

```
# i  
c(  
  complete_pooling = sigma(m_complete),  
  no_pooling = sigma(m_nopool),  
  partial_pooling = sigma(m_partial)  
)
```



```
## complete_pooling      no_pooling  partial_pooling  
##           24.09055           11.70741           11.70740
```


don't remove the missing values – this is also information about the data

```
# clean data and fit model  
dat <- politeness_1 %>% drop_na(f0mn, total_duration, subject)  
m_subject <- lmer(total_duration ~ f0mn + (1 | subject), data = dat)  
  
# add fitted values to the dataset  
dat$fit <- fitted(m_subject)
```

include estimates here not just free text

- 2) Making the second-level effects more complicated - now fit the two models with the formulas: `f0mn ~ attitude * gender + (1 | subject) + (1 | scenario)` and `f0mn ~ attitude * gender + (gender | subject) + (1 | scenario)`

```
mA <- lmer(f0mn ~ attitude * gender + (1 | subject) + (1 | scenario),  
           data = politeness_1)  
  
mB <- lmer(f0mn ~ attitude * gender + (gender | subject) + (1 | scenario),  
           data = politeness_1)
```

- i. Explain what each of the group-level effects and `_subject_` and `_scenario_` effects mean

Please use this

Melanie (MV)

- iii. how would you model a no-pooling model that took both *subject* and *scenario* into account?

Melanie (MV)

+ **ii. now fit a similar model, but this time have *scenario* be a nominal variable**